Executive Summary

Retail Analytics Data Ingestion & Transformation Pipeline

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Course: Introduction to Python Programming

The Retail Analytics Data Ingestion and Transformation Pipeline was developed to automate the validation and analysis of retail transaction data. The system imports raw transaction records, validates required fields, rejects invalid transactions, and transforms data into useful business metrics.

The project was designed using a modular architecture with separate Python modules responsible for validation, transformation, reporting, and entity modeling. The TransactionRecord class was used to represent retail transactions, while additional functionality was organized into reusable modules.

Functional programming techniques such as map(), filter(), and lambda expressions were used to perform stateless data transformations. Error handling was implemented using try/except blocks to improve reliability and data integrity.

The application generates key metrics including total sales, top-performing branch, customer tier distribution, and transaction summaries. Unit tests were also created to verify validation functionality.

This project demonstrates procedural programming, object-oriented programming, modular software design, functional programming, and data validation techniques learned during the course.